

Loading Data



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Overview



Load data in CSV format

Use the options & modes for writing

Understand how data is saved as file

Manage Spark's partitioning

Load data in Apache Parquet format

Understand databases & tables

Work with managed & unmanaged tables

Load data in tables



Loading to Files



Why Multiple Files Are Created?



RDDs/DataFrames are broken into partitions, and partitions are distributed to the memory of multiple nodes in a cluster



Each node can have multiple partitions, and processing happens on partitions – helping to achieve parallelism



While saving, a folder is created instead of file, and each partition is written as a separate file inside the folder



Default number of partitions is 200, so 200 part files are created inside the folder



DataFrame Partitioning



Specify folder name to read all the partitioned files, using Spark jobs or tools with Spark connector



Default partitions can be changed using `spark.sql.shuffle.partitions` configuration setting



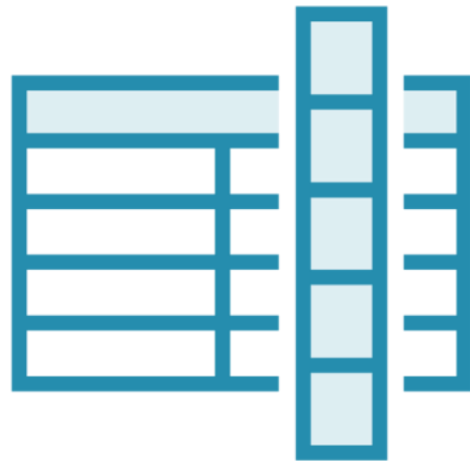
Use `coalesce` or `repartition` methods to change partitions on a DataFrame. Changing result in movement of data between partitions



Rule of thumb – use `repartition()` to increase partitions, and `coalesce()` to decrease partitions



Apache Parquet



Columnar storage format

Support complex nested data structures

Stores schema in the file itself

Supports efficient compression & encoding

Binary files

Writing to Parquet is slower

Querying is much faster than CSV/JSON

Working with Databricks Tables



Databases and Tables

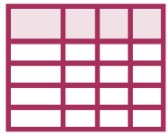


Table is a representation of dataset stored at an external location, and database contain tables



Each workspace has central Hive metastore, to store the metadata of tables, which is accessible by all the clusters



Underlying data can be in any format – CSV, JSON, Parquet, RDBMS table etc.



Once defined, use it as a table in SQL, or as a DataFrame in Scala, without providing credentials



Table Types

Managed Table

Schema and data is managed by Spark

Data is stored in DBFS

Stored in Parquet format

Dropping table deletes both the
schema and the data

Useful to persist staging data

Unmanaged Table

Only schema is managed by Spark

Data is stored in an external location

Stored in underlying dataset format

Dropping table deletes only the
schema, data is not deleted from source

Useful to persist processed data



Summary



Load data in CSV format and use options

- header, dateFormat, SaveMode...

Multiple partition files created while saving

Manage partitioning

- spark.sql.shuffle.partitions config
- coalesce, repartition methods

Load data in Parquet, compared with CSV

- Higher read performance, compression

Databricks support Databases & Tables

- Metadata is registered in Hive metastore

Managed vs. Unmanaged tables

